Shammas Ahmed

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Education

Rice University Houston, TX

B.S. Operations Research (specialization in Algorithms) *Expected May 2027*

- GPA: 3.927/4.000
- **Mathematics:** Honors Linear Algebra, Real Analysis I, Probability and Statistics for Data Science, Geometry
- **Optimization & Algorithms:** Linear and Integer Programming, Large-Scale Optimization, Algorithm Design
- **Other:** Machine Learning with Graphs, Simulation Modeling and Analysis
- **Graduate-level courses:** Convex Optimization, Discrete Optimization, Online Optimization, Stochastic Models and Optimization

Research Interests

I am interested in the design and analysis of algorithms for discrete optimization, with a focus on graph and network problems, including shortest-path variants, network flows, and questions of correctness, hardness and approximation. I am also interested in designing algorithms for optimization under uncertainty, particularly in characterizing optimal policies for competing agents in online selection problems such as the secretary problem and Pandora's box. More broadly, I enjoy optimization problems that can be modeled as a game.

Research Experience

DSO Summer Scholar: Decision-Focused Learning May 2026 – present

Data Science & Operations Department, USC Marshall *Los Angeles, CA*

Advisor(s): Prof. Vishal Gupta and Prof. Kabir Verchand

- Analyzed benchmarking and performance metrics across 24 papers on machine learning for optimization
- Established a constant additive "look-ahead premium" (LAP) in regret-based evaluation that distorts comparisons of model performance and benchmark difficulty
- Proposed a regret-reporting policy that accounts for LAP to reflect true model performance on given instances
- Research manuscript accepted at the NeurIPS MLxOR Workshop 2026
- Expanding the 24-paper audit to 300 papers for a journal manuscript

Undergraduate Research Assistant: Mixed-Integer Programming May 2025 – Dec 2025

Operations Research Department, Rice University *Houston, TX*

Advisor(s): Prof. Sebastian Perez-Salazar

- Investigated a shortest-path problem with homology constraints (applications in robotics and healthcare)
- Reformulated the problem as a network-flow optimization model and conjectured NP-hardness
- Developed a polynomial-time heuristic and characterized instances for which the performance ratio is finite
- Implemented the model in Python and Gurobi, using flow decompositions and harmonic projection constraints on 20×20 grids, and visualized solutions

Summer Undergraduate Research Fellow: Inverse Optimization May 2024 – August 2024

Operations Research Department, Rice University *Houston, TX*

Advisor(s): Prof. Fatemeh Nosrat

- Studied the inverse integer knapsack problem, analyzed geometric structure and characterized "inverse feasible regions" for different feasible solutions
- Independently developed closed-form solution for optimal inverse cost vectors that matches existing literature
- Presented findings at the Rice Summer Undergraduate Research Symposium and the UT Arlington Math Symposium

Publications

S. Ahmed, Vishal Gupta. **Hindsight is 20/20: Making Regret Meaningful in Decision-Focused Learning**. *Accepted, NeurIPS Workshop on Machine Learning for Operations Research (MLxOR), 2026.*

S. Ahmed, Sebastian Perez-Salazar. **Optimal Path Planning with Soft Homology Constraints**. *Manuscript, available on personal website.*

Presentations

Hindsight is 20/20: Making Regret Meaningful in Decision-Focused Learning
NeurIPS MLxOR Workshop — Dec. 2026 (forthcoming)

Regret Metrics in Decision-Focused Learning
DSO Summer Research Showcase — Aug. 2026

The Inverse Integer Knapsack Problem
UT Arlington Math Symposium — Nov. 2024

The Inverse Integer Knapsack Problem
Rice Summer Undergraduate Research Symposium — Aug. 2024

Selected Projects

Competitive Pandora's Box | *Course project, CMOR 646 – Online Optimization* — Spring 2026
- Modeled a two-agent variant of Pandora's box and showed that Weitzman's policy is suboptimal under competition
- Derived a stopping threshold under which players should stop opening boxes earlier than under Weitzman's policy

Wordle as a Graph Problem | *Course project, COMP 459 – Machine Learning with Graphs* — Spring 2026
- Implemented a Wordle solver that combines multiple linguistic corpora using graph-based label spreading and a neural network gating system to address the game as a "weighted experts" problem

Remodeling ExxonMobil's Caustic Soda Supply Chain | *Course project, CMOR 492 – Senior Design* — Fall 2026
- Advising ExxonMobil on whether caustic soda from global suppliers can be sourced cost-effectively for North American plants by restructuring their supply chain
- Sourcing global pricing data on caustic and transportation, modeling as a single-commodity flow problem

The Bike Rebalancing Problem | *Course project, CMOR 451 – Simulation Modeling & Analysis* — Fall 2026
- Simulating bike sharing using NYC's CitiBike data and determining the optimal bike rebalancing policy
- Characterizing long-run behavior and confirming optimal policy on Boston's BlueBike data

Teaching Experience

Teaching Assistant: Introduction to Engineering Computation — August 2024 – present
Rice University School of Engineering and Computing — *Houston, TX*
- Deliver weekly 1-hour lectures on concepts in math and engineering (e.g., neural networks, first-order optimization methods, root-finding methods) with implementation in MATLAB and Python
- Lead weekly office hours, grade assignments, and give feedback to 20+ students

Teaching Assistant: Reasoning about Algorithms — August 2026 – present
Rice University School of Engineering and Computing — *Houston, TX*
- Lead weekly office hours and grade assignments on algorithm design, runtime analysis and correctness
- Proctor labs and advise instructors weekly on assignment rubrics and grading methods

Honors, Awards & Scholarships

DSO Summer Scholar – USC Marshall — 2026
INFORMS Annual Meeting Undergraduate Travel Scholarship (Alternate) – INFORMS — 2026
Continuing Research Excellence Student – Rice University — 2025
Summer Undergraduate Research Fellow – Rice University — 2024
President's Honor Roll – Rice University — Fall 2023 – Fall 2024, Spring 2026

Service & Leadership

New Student Orientation Week Coordinator — December 2024 – August 2025
Rice University — *Houston, TX*
- Directed 9-month planning process of week-long, campus-wide orientation for 1,200 incoming students
- Conducted 80+ hours of volunteer interviews, led volunteer trainings, and secured $5k+ in sponsorships

Prospective Student Interviewer — August 2026 – present
Rice University — *Houston, TX*
- Interview 75+ prospective new students and represent Rice to applicants
- Communicate with the admissions office on student performance, with recommendations on final decisions

Skills

Programming: Python, MATLAB, R
Optimization: Gurobi, PyEPO, PuLP
Tools: Git, GitHub, VS Code, Google Colab
Methods: Mathematical modeling, algorithm design, data analysis, data visualization